PROJECT REPORT

CliniSence-AI: An AI-Powered Clinical Intelligence & Patient Assistance Platform Using React, Flask, MongoDB and OCR

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Course Code: SCAS002 – Data Science

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Acknowledgement

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This project has enhanced my practical understanding of Artificial Intelligence, Natural Language Processing, OCR, Full Stack Development, REST API development, and healthcare software engineering.

Contents

Acknowledgement 1

1 Objective of the Project 3

2 Tools & Technologies Used 3

3 Areas Covered 3

4 Title ..... **Error! Bookmark not defined.**

5 Problem Definition 4

6 Scope and Objectives 4

7 System Requirements 5

8 Methodology 5

9 Modules 5

10 Sample Code Snippet **7**
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11 Output 8

12 Challenges Faced 8

13 Learnings 9

14 Summary 9

1 Objective of the Project

The primary objective of **CliniSense AI** is to develop an intelligent healthcare platform that enables users to upload medical reports and receive AI-assisted clinical analysis in an easy-to-understand format.

The system extracts text from uploaded medical reports using Optical Character Recognition (OCR), analyses laboratory parameters through a clinical intelligence engine, generates patient-friendly summaries, estimates health scores, identifies potential health risks, and provides appropriate medical recommendations.

Unlike conventional healthcare systems that merely store reports, CliniSense AI focuses on assisting patients in understanding their reports while supporting healthcare professionals with structured clinical insights.

The project integrates Artificial Intelligence, OCR, Natural Language Processing, PDF generation, REST APIs, and Full Stack Development into one scalable healthcare platform.

Training Overview

2 Tools & Technologies Used

- **Programming Language**: JavaScript (ES6+), Python, HTML5, CSS3
- **Development Environment**: Visual Studio Code, MongoDB, Postman, GitHub
- **Frontend Technologies**: React.js, Tailwind CSS, AXIOS
- **Core Subject Focus**: Database Management (SQL)

3 Areas Covered

- Artificial Intelligence
- Healthcare Analytics
- Optical Character Recognition (OCR)
- Natural Language Processing (NLP)
- Full Stack Development
- REST API Development
- Database Management

- Authentication & Authorization
- Clinical Intelligence

4 Problem Definition

Medical reports often contain complex laboratory values, abbreviations, and technical terminology that are difficult for patients to understand without professional assistance.

Traditional healthcare systems mainly focus on storing patient reports but rarely provide simplified explanations or intelligent analysis.

Patients frequently rely on healthcare professionals even for understanding routine laboratory reports, resulting in delays and confusion.

CliniSense AI addresses these challenges by integrating OCR and Artificial Intelligence to automatically analyze uploaded medical reports, identify abnormal parameters, generate patient-friendly summaries, calculate health scores, estimate health risks, and provide recommendations in a structured format.

5 Scope and Objectives

Develop a secure healthcare platform for patients.

- Implement secure user registration and JWT authentication.
- Allow uploading of PDF and image medical reports.
- Extract medical text using OCR.
- Analyze medical parameters using an AI-powered clinical reasoning engine.
- Calculate Health Score.
- Predict Risk Level.
- Generate AI-powered summaries.
- Generate downloadable PDF reports.
- Store complete report history in MongoDB.
- Provide a responsive dashboard for users.

6 System Requirements

- **Platform**: Any modern web browser (Chrome, Firefox, Edge)
- **Language**: JavaScript, HTML, CSS, Python, SQL
- **RAM**: 8 GB / 512MB minimum
- **Operating System**: Windows 10/11, Linux, or macOS

7 Methodology

- The system begins with secure user registration and authentication using JWT.
- After login, users upload medical reports in PDF or image format.
- The uploaded report is processed using Tesseract OCR to extract medical text.
- The extracted text undergoes Natural Language Processing for keyword detection and parameter extraction.
- The Clinical Intelligence Engine analyzes laboratory parameters, compares them against predefined normal ranges, calculates a Health Score, estimates Risk Level, generates findings and recommendations, and produces a patient-friendly summary.
- The complete analysis is stored in MongoDB.
- Users can review previous reports, visualize report details, and download a professional AI-generated PDF report.

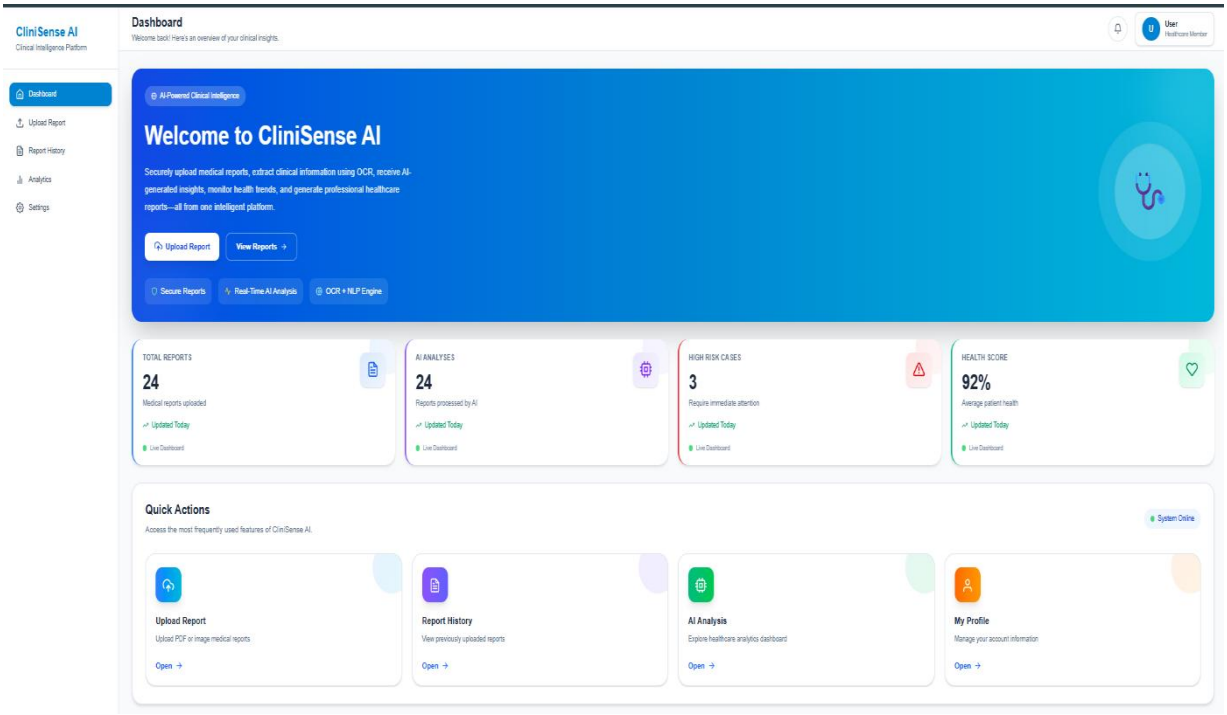
8 Modules

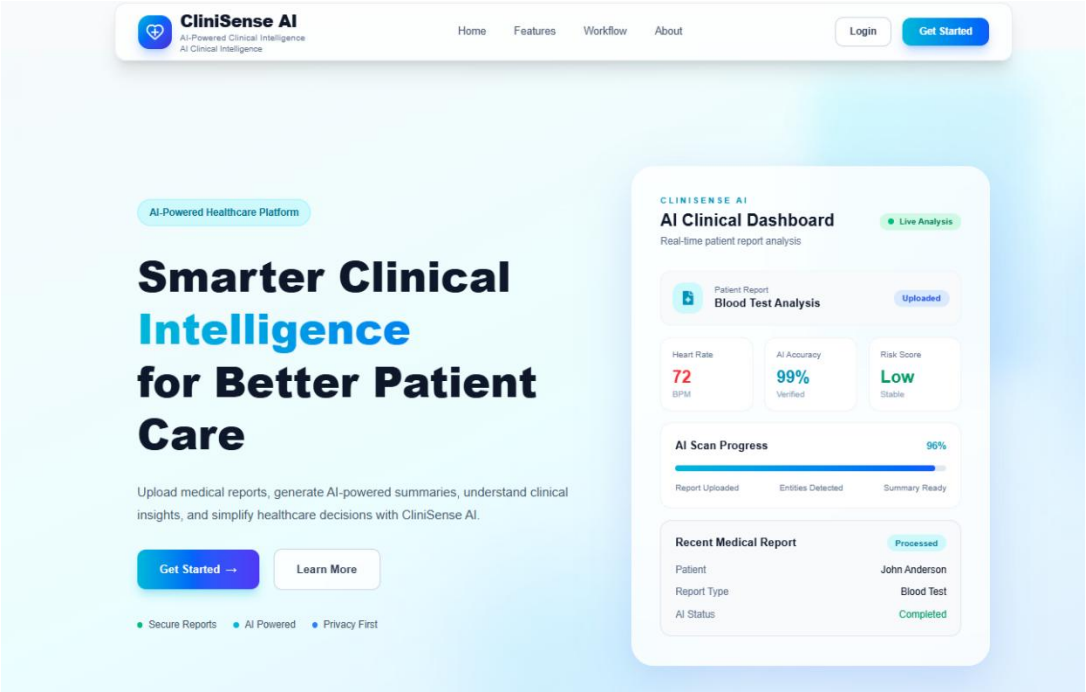
1. **User Authentication Module:** Enables secure user registration, login, logout, and account management using **JWT (JSON Web Token)** authentication and encrypted passwords. This module ensures that only authorized users can access their medical reports and personalized healthcare dashboard.

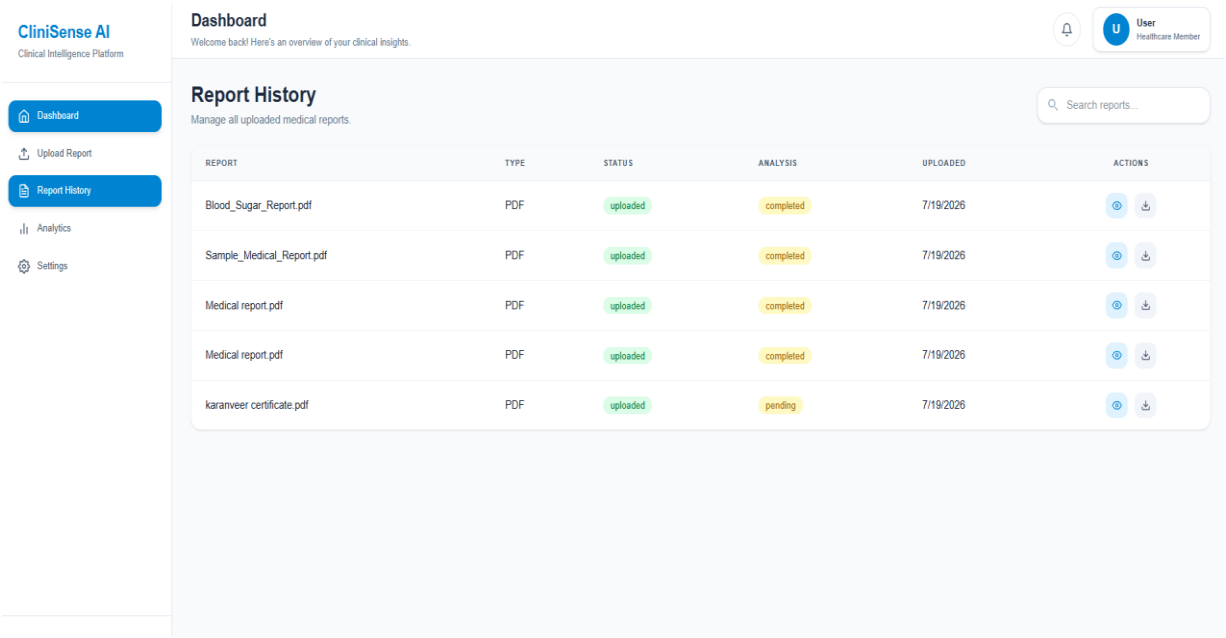
2. **User Profile & Career Module:** Allows users to upload medical reports in **PDF and image formats** through a secure interface. The module validates uploaded files, stores them securely on the server, and maintains report metadata in the MongoDB database for future access and analysis.**AI Study Planner Module:** Generates personalized daily, weekly, and monthly study plans along with learning timelines to help users achieve their academic objectives.

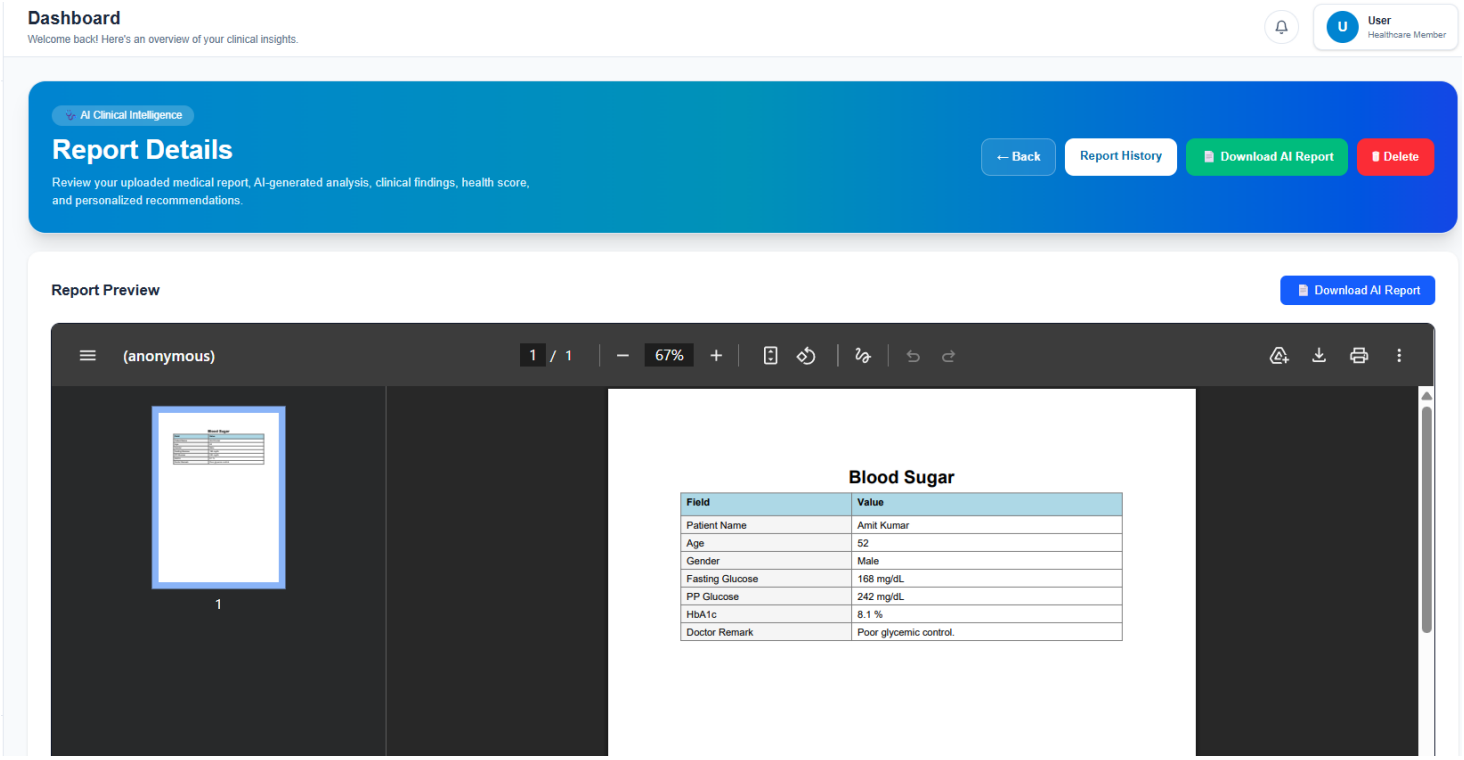
3. **Task & Progress Management Module:** Extracts medical text from uploaded reports using **Tesseract OCR** and image processing techniques. The module converts scanned PDFs and images into machine-readable text, forming the basis for further AI-powered clinical analysis.

Screenshot: -


CliniSense AI
Clinical Intelligence Platform
Dashboard
Welcome back! Here's an overview of your clinical insights.
Dashboard
Upload Report
Report History
Analytics
Settings
AI-Powered Clinical Intelligence
Welcome to CliniSense AI
Securely upload medical reports, extract clinical information using OCR, receive AI-generated insights, monitor health trends, and generate professional healthcare reports—all from one intelligent platform.
Upload Report
View Reports
Secure Reports
Real-Time AI Analysis
OCR + NLP Engine
TOTAL REPORTS
24
Medical reports uploaded
Updated Today
Live Dashboard
AI ANALYSES
24
Reports processed by AI
Updated Today
Live Dashboard
HIGH RISK CASES
3
Require immediate attention
Updated Today
Live Dashboard
HEALTH SCORE
92%
Average patient health
Updated Today
Live Dashboard
Quick Actions
Access the most frequently used features of CliniSense AI.
System Online
Upload Report
Upload PDF or image medical reports
Open
Report History
View previously uploaded reports
Open
AI Analysis
Explore healthcare analytics dashboard
Open
My Profile
Manage your account information
Open


CliniSense AI
AI-Powered Clinical Intelligence
AI Clinical Intelligence
Home
Features
Workflow
About
Login
Get Started
AI-Powered Healthcare Platform
Smarter Clinical Intelligence for Better Patient Care
Upload medical reports, generate AI-powered summaries, understand clinical insights, and simplify healthcare decisions with CliniSense AI.
Get Started →
Learn More
Secure Reports
AI Powered
Privacy First
CLINISENSE AI
AI Clinical Dashboard
Real-time patient report analysis
Live Analysis
Patient Report
Blood Test Analysis
Uploaded
Heart Rate
72
BPM
AI Accuracy
99%
Verified
Risk Score
Low
Stable
AI Scan Progress
96%
Report Uploaded
Entities Detected
Summary Ready
Recent Medical Report
Processed
Patient
John Anderson
Report Type
Blood Test
AI Status
Completed


CliniSense AI
Clinical Intelligence Platform
Dashboard
Upload Report
Report History
Analytics
Settings
Dashboard
Welcome back! Here's an overview of your clinical insights.
User
Healthcare Member
Report History
Manage all uploaded medical reports.
Search reports...
REPORT
TYPE
STATUS
ANALYSIS
UPLOADED
ACTIONS
Blood_Sugar_Report.pdf
PDF
uploaded
completed
7/19/2026
Sample_Medical_Report.pdf
PDF
uploaded
completed
7/19/2026
Medical report.pdf
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uploaded
completed
7/19/2026
karanveer certificate.pdf
PDF
uploaded
pending
7/19/2026


Dashboard
Welcome back! Here's an overview of your clinical insights.
User
Healthcare Member
AI Clinical Intelligence
Report Details
Review your uploaded medical report, AI-generated analysis, clinical findings, health score, and personalized recommendations.
← Back
Report History
Download AI Report
Delete
Report Preview
Download AI Report
(anonymous)
1 / 1
67%
1
Blood Sugar
Field
Value
Patient Name
Amit Kumar
Age
52
Gender
Male
Fasting Glucose
168 mg/dL
PP Glucose
242 mg/dL
HbA1c
8.1 %
Doctor Remark
Poor glycemic control.

Results and Discussion

9 Output

- Users can securely **register, log in, and manage their accounts** through an interactive web interface protected with **JWT authentication** and encrypted passwords.
- Users can upload **medical reports in PDF and image formats**, which are securely stored and processed for AI-based clinical analysis.
- The **AI Clinical Intelligence Engine** analyzes the extracted data, identifies abnormal medical parameters, calculates the **Health Score**, predicts the **Risk Level**, and generates patient-friendly summaries along with personalized health recommendations.
- Users can view their **complete report history**, access detailed AI analysis, preview uploaded reports, and download professionally formatted **AI-generated PDF reports** containing clinical findings, health scores, and recommendations.
- All user information, uploaded reports, AI analysis results, and report history are securely stored and managed in the **MongoDB database**, enabling reliable data management and future healthcare analysis.

10 Challenges Faced

- **OCR Integration:** Extracting accurate medical text from different PDF and image report formats using Tesseract OCR, while handling variations in report layouts, image quality, and scanned documents**.**
- **AI Clinical Analysis:** Developing a rule-based clinical intelligence engine capable of identifying medical parameters, comparing them with standard reference ranges, calculating health scores, determining risk levels, and generating meaningful patient-friendly summaries and recommendations**.**
- **System Integration:** Maintaining smooth communication between the React frontend, Flask backend, OCR processing module, AI Clinical Intelligence Engine,

and MongoDB database while ensuring secure JWT authentication and real-time report processing**.**

11 Learnings

- Gained practical experience in developing a **full-stack healthcare application** using **React.js, Flask, MongoDB, and Python**, while implementing secure user authentication and RESTful API communication.
- Learned to integrate **Optical Character Recognition (OCR)** and **Artificial Intelligence** techniques for extracting medical information, analyzing laboratory reports, and generating patient-friendly clinical summaries and recommendations.
- Improved understanding of **MongoDB database design**, CRUD operations, REST API development, JWT authentication, and secure management of medical reports and user information.
- Enhanced problem-solving, debugging, and software development skills while building a **scalable, secure, and user-friendly AI-powered clinical intelligence platform** for healthcare applications.

12 Summary

CliniSense AI successfully demonstrates the integration of **Artificial Intelligence, Optical Character Recognition (OCR), Natural Language Processing (NLP), and Full-Stack Web Development** to create an intelligent healthcare platform for medical report analysis. The system enables users to securely upload medical reports, extract clinical information using OCR, perform AI-assisted analysis, calculate health scores, predict risk levels, generate patient-friendly summaries, and download professional AI-generated PDF reports.

The application ensures secure user authentication, efficient report management, complete CRUD operations, and reliable data storage using **MongoDB**, making it scalable, secure, and user-friendly. By combining **React.js, Flask, MongoDB, Tesseract OCR, Report Lab, and AI-based clinical intelligence**, CliniSense AI provides a modern healthcare solution that improves patient understanding of medical reports and supports informed healthcare decisions. This project has strengthened my understanding of **Artificial Intelligence, OCR, Natural Language Processing, Full-Stack Development, REST API development, database management, and healthcare software engineering**, making it a valuable contribution to the field of intelligent healthcare